

The finiteness theorem

SGA 2, Exposé VIII

English draft

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About this draft

This is an unofficial English translation of Exposé VIII, “The finiteness theorem”, from SGA 2. The source volume is A. Grothendieck (notes by a group of auditors), with an exposé by M. Raynaud, *Cohomologie locale des faisceaux cohérents et théorèmes de Lefschetz locaux et globaux* (SGA 2), Séminaire de géométrie algébrique du Bois Marie, 1962. It follows the slightly corrected SMF recomposition, arXiv:math/0511279.

This draft contains the assigned body of the source, including proofs, original footnotes, and editor notes (N.D.E.). Statement numbering and mathematical notation follow the source. References to material outside this draft retain their original numbers. Apparent mathematical misprints in the source have been retained; they are recorded in the accompanying README, separately from the translated text. Scholarly proofreading remains outstanding.

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EXPOSÉ VIII

The finiteness theorem

1. A spectral sequence of biduality¹

Let us state the result which we wish to reach:

PROPOSITION 1.1. *Let A be a noetherian ring and let I be an ideal of A . Set $X = \text{Spec}(A)$ and $Y = V(I)$. Let M be an A -module of finite type and of finite projective dimension. Let $F = \widetilde{M}$ be the $\mathcal{O}_X$ -Module associated with M .*

1) *There exists a spectral sequence:*

$$H_Y(X, F) \Leftarrow \text{Ext}_Y^p(\text{Ext}^{-q}(M, A), A).$$

2) *There exists a spectral sequence:*

$$\underline{H}_Y(X, F) \Leftarrow \underline{\text{Ext}}_Y^p(\underline{\text{Ext}}^{-q}(F, \mathcal{O}_X), \mathcal{O}_X).$$

Of course, 2) is deduced from 1) by remarking that, if M and N are two A -modules of finite type and if one sets $F = \widetilde{M}$ and $G = \widetilde{N}$, one has isomorphisms:

$$\begin{aligned} \underline{H}_Y(F) &\approx \widetilde{H_Y(X, F)}, \\ \underline{\text{Ext}}_Y(F, G) &\approx \widetilde{\text{Ext}_Y(F, G)}, \\ \underline{\text{Ext}}_{\mathcal{O}_X}(F, G) &\approx \widetilde{\text{Ext}_A(M, N)}. \end{aligned}$$

Let $\underline{\mathbb{C}}$ be the category of A -modules, and $\underline{\text{Ab}}$ that of abelian groups. Let $\underline{F}$ be the functor:

$$\begin{aligned} \underline{F} : \underline{\mathbb{C}} &\longrightarrow \underline{\text{Ab}} \quad \text{defined by} \\ M &\longmapsto \Gamma_Y(\widetilde{M}). \end{aligned}$$

One knows from Exposé II that there exists an isomorphism of ∂ -functors:

$$H_Y^*(X, \widetilde{M}) \approx R^* \underline{F}(M).$$

Moreover, let Ext_Y^* be the right derived functors in the second argument of

$$\underline{F} \circ \text{Hom} : \underline{\mathbb{C}}^\circ \times \underline{\mathbb{C}} \longrightarrow \underline{\text{Ab}}.$$

One knows from Exposé VI that one has an isomorphism of ∂ -functors:

$$\text{Ext}_Y^*(M, N) \simeq \text{Ext}_Y^*(\widetilde{M}, \widetilde{N}).$$

Finally, let us retain the following result from Exposé VI: if C is an injective A -module and if N is an A -module of finite type, the sheaf $\underline{\text{Hom}}(\widetilde{N}, \widetilde{C}) \approx \widetilde{\text{Hom}(N, C)}$ is *flasque*, hence $R^1 \underline{F}(\text{Hom}(N, C)) = 0$.

It remains for us to prove the following result:

LEMMA 1.2. *Let A be a noetherian ring and let $\underline{\mathbb{C}}$ be the category of A -modules. Let $\underline{F} : \underline{\mathbb{C}} \rightarrow \underline{\text{Ab}}$ be a left exact additive functor such that, for every A -module N of finite type and every injective A -module C , one has $R^1 \underline{F}(\text{Hom}(N, C)) = 0$. Let M be an A -module of finite type and of finite projective dimension. There exists a spectral sequence:*

$$R^* \underline{F}(M) \Leftarrow \text{Ext}_{\underline{F}}^p(\text{Ext}^{-q}(M, A), A),$$

¹The reader familiar with the language of Verdier's derived categories will recognize the spectral sequence associated with a *biduality isomorphism*. Cf. SGA 6 I.

where $\text{Ext}_{\mathbb{F}}^p$ denotes the p -th right derived functor of $\mathbb{F} \circ \text{Hom}$.

We shall consider only complexes whose differential is of degree $+1$. By the hypothesis made on M , there exists a projective resolution of M of finite length:

$$u: L^\bullet \longrightarrow M,$$

where, moreover, the L^p are modules of finite type and $L^p = 0$ if $p \notin [-n, 0]$. Let moreover $v: M \rightarrow I^\bullet$ be an injective resolution of M . I say that

$$(1.1) \quad v \circ u: L^\bullet \longrightarrow I^\bullet$$

is an injective resolution of $L^\bullet$. It is appropriate to make precise what this means.

DEFINITION 1.3. Let $X^\bullet$ be a complex of A -modules; one calls an injective resolution of $X^\bullet$ a homomorphism of complexes:

$$x: X^\bullet \longrightarrow CX^\bullet,$$

such that CX^p is injective for every $p \in \mathbb{Z}$, and such that x induces an isomorphism in homology.

PROPOSITION 1.4. Every complex bounded on the left, i.e. such that there exists $q \in \mathbb{Z}$ with $X^p = 0$ for $p < q$, admits an injective resolution. Moreover, if $u: X^\bullet \rightarrow Y^\bullet$ is a homomorphism of complexes (bounded on the left) and if $x: X^\bullet \rightarrow CX^\bullet$ and $y: Y^\bullet \rightarrow CY^\bullet$ are injective resolutions of $X^\bullet$ and $Y^\bullet$, there exists a homomorphism of complexes:

$$Cu: CX^\bullet \longrightarrow CY^\bullet,$$

unique up to homotopy, such that the diagram:

$$\begin{array}{ccc} X^\bullet & \xrightarrow{x} & CX^\bullet \\ u \downarrow & & \downarrow Cu \\ Y^\bullet & \xrightarrow{y} & CY^\bullet \end{array}$$

is commutative up to homotopy.

The proof is left to the reader².

Let us recall a notation introduced in Exposé V.

Notation. Let $X^\bullet$ and $Y^\bullet$ be two complexes. One denotes by $\text{Hom}^\bullet(X^\bullet, Y^\bullet)$ the simple complex whose component of degree n is

$$(\text{Hom}^\bullet(X^\bullet, Y^\bullet))^n = \prod_{-p+q=n} \text{Hom}(X^p, Y^q)$$

also denoted $\text{Hom}^n(X^\bullet, Y^\bullet)$, and whose differential is given by:

$$\begin{aligned} \partial_n: \text{Hom}^n(X^\bullet, Y^\bullet) &\longrightarrow \text{Hom}^{n+1}(X^\bullet, Y^\bullet) \\ \partial_n &= d' + (-1)^{n+1}d'', \end{aligned}$$

where d' and d'' are the differentials (of degree $+1$) induced by those of $X^\bullet$ and of $Y^\bullet$.

Now let $A^\bullet$ be the complex defined by $A^p = 0$ if $p \neq 0$ and $A^0 = A$. Let

$$a: A^\bullet \longrightarrow CA^\bullet$$

be an injective resolution of $A^\bullet$. Consider the double complex:

$$(1.2) \quad Q^{p,q} = \text{Hom}(\text{Hom}(L^{-q}, A), CA^p).$$

The first spectral sequence of the bicomplex $\mathbb{F}Q^{\bullet\bullet}$ will give the conclusion of Lemma 1.2.

Set

$$(1.3) \quad L'^\bullet = \text{Hom}^\bullet(L^\bullet, A^\bullet),$$

²Cf. also H. Cartan & S. Eilenberg, *Homological Algebra*, Princeton Math. Series, vol. 19, Princeton University Press, 1956.

and

$$(1.4) \quad P^\bullet = \text{Hom}^\bullet(L'^\bullet, CA^\bullet).$$

One sees easily that $P^\bullet$ is the simple complex associated with $Q^{\bullet\bullet}$. Let us compute the abutment of the spectral sequence i.e. the homology of $\underline{F}P^\bullet$. For this, using the fact that $L^\bullet$ is projective of finite type in every dimension, one proves that $L^\bullet$ is isomorphic to $\text{Hom}^\bullet(L'^\bullet, A^\bullet)$. From the homomorphism $a: A^\bullet \rightarrow CA^\bullet$, one deduces a homomorphism:

$$b: \text{Hom}^\bullet(L'^\bullet, A^\bullet) \longrightarrow \text{Hom}^\bullet(L'^\bullet, CA^\bullet),$$

or again, a homomorphism

$$(1.5) \quad c: L^\bullet \longrightarrow P^\bullet.$$

This being said, it is easy to see, using the fact that $L'^\bullet$ is projective of finite type in every dimension and bounded on the left, that (1.5) is an injective resolution of $L^\bullet$. Using Proposition 1.4., one concludes that $P^\bullet$ is homotopically equivalent to $I^\bullet$, where $I^\bullet$ is the injective resolution of M introduced above (1.1). One deduces that the *abutment* of the first spectral sequence of the double complex $\underline{F}Q^{\bullet\bullet}$, which is $H^*(\underline{F}P^\bullet)$, is isomorphic to $R^*\underline{F}(M)$.

The initial term of the first spectral sequence of the bicomplex $\underline{F}Q^{\bullet\bullet}$ is:

$$E_2^{p,q} = {}'H^p({}''H^q(\underline{F}Q^{\bullet\bullet})).$$

For every $p \in \mathbf{Z}$, CA^p is injective. By the hypothesis made on $\underline{F}$, the functor (restricted to the category of modules of finite type):

$$N \longmapsto \underline{F}\text{Hom}(N, CA^p)$$

is exact. Whence one deduces isomorphisms:

$${}''H^q(\underline{F}\text{Hom}(L'^\bullet, CA^p)) \approx \underline{F}\text{Hom}(H^{-q}(L'^\bullet), CA^p).$$

By the definition of $\text{Ext}_{\underline{F}}^*$ as derived functor of $\underline{F} \circ \text{Hom}$, one deduces isomorphisms:

$$E_2^{p,q} \approx \text{Ext}_{\underline{F}}^p(H^q(L'^\bullet), A).$$

Now $L'^\bullet = \text{Hom}^\bullet(L^\bullet, A^\bullet)$, where $L^\bullet$ is a projective resolution of M whence isomorphisms:

$$\text{Ext}^q(M, A) \approx H^q(L'^\bullet),$$

which gives the conclusion. $\square$

2. The finiteness theorem

THEOREM 2.1³. *Let X be a locally noetherian prescheme, Y a closed subset of X and $\underline{F}$ a coherent $\mathcal{O}_X$ -Module. Suppose that X is locally immersible in a regular prescheme⁴. Let $i \in \mathbf{Z}$. Suppose that:*

a) *for every $x \in U = X - Y$, one has*

$$H^{i-c(x)}(\underline{F}_x) = 0,$$

where one has set⁵:

$$(2.6) \quad c(x) = \text{codim}(\overline{\{x\}} \cap Y, \overline{\{x\}}).$$

Then:

b) $\underline{H}_Y^i(\underline{F})$ *is coherent.*

³**N.D.E.** for an analogous statement, but in a slightly more general situation, see Mme Raynaud (Raynaud M., “Théorèmes de Lefschetz en cohomologie des faisceaux cohérents et en cohomologie étale. Application au groupe fondamental”, *Ann. Sci. Éc. Norm. Sup. (4)* **7** (1974), p. 29–52, proposition II.2.1).

⁴This condition may be generalized to the hypothesis of the existence, locally on X , of a “dualizing complex”, in the sense defined in R. Hartshorne, *Residues and duality* (cited in the note (cithar) of Exp. IV p. pagecithar).

⁵**N.D.E.** as in Exposé V, $H^*(\underline{F}_x)$ denotes the local cohomology $H_{\mathfrak{m}_x}(\underline{F}_x)$.

COROLLARY 2.2⁶. Under the hypotheses of the preceding theorem, condition a) is equivalent to:

c) for every $x \in U$ such that $c(x) = 1$, one has $H^{i-1}(F_x) = 0$.

COROLLARY 2.3. Let X be a locally noetherian prescheme locally immersible in a regular prescheme, Y a closed subset of X , F a coherent $\mathcal{O}_X$ -Module, n an integer. The following conditions are equivalent:

- (i) for every $x \in U$, one has $\text{prof } F_x > n - c(x)$;
- (ii) for every $x \in U$ such that $c(x) = 1$, one has $\text{prof } F_x \geq n$;
- (iii) for every $i \in \mathbf{Z}$, $H_Y^i(F)$ is coherent if $i \leq n$.
- (iv) $R^i i_*(F|_U)$ is coherent for $i < n$ ⁷.

Suppose these results acquired when X is the spectrum of a regular noetherian ring A and when F is the sheaf associated to an A -module of finite projective dimension.

Let us first remark that, if $(X_j)_{j \in J}$ is an open covering of X by opens immersible in a regular scheme, each of the conditions above is equivalent to the conjunction of the analogous conditions obtained by replacing X by X_j , Y by $Y_j = Y \cap X_j$ and F by $F|_{X_j}$. Indeed, only the conditions involving $c(x)$ can present a difficulty. Let $x \in U$. If $x \in X_j$, set

$$c_j(x) = \text{codim}(X_j \cap \overline{\{x\}} \cap Y, X_j \cap \overline{\{x\}}),$$

one has necessarily $c_j(x) \geq c(x)$. Let $y \in \overline{\{x\}} \cap Y$, which “gives the codimension”, i.e. such that $c(x) = \dim \mathcal{O}_{\overline{\{x\}}, y}$, let X_j be an open of the covering such that $y \in X_j$, then $x \in X_j$, hence $c_j(x) = c(x)$, which allows one to conclude that a) for the X_j implies a) for X . At this stage, one has only a partial converse, namely that a) for X implies a) for those X_j such that $c(x) = c_j(x)$, which suffices for our purpose.⁸

One chooses a covering of X by opens immersible in a regular prescheme. Applying the preceding, one sees that one may suppose X closed in a regular X' . The reduction to X' is then immediate.

One may therefore suppose X regular, and even affine by covering X by affine opens. That one may suppose that $F = \widetilde{M}$, where M is of finite projective dimension, will result from the following lemma:

LEMMA 2.4. Let X be a regular noetherian prescheme. Let F be a coherent $\mathcal{O}_X$ -Module. The function which to every $x \in X$ associates the projective dimension of F_x is bounded above.

Indeed, let $x \in X$ and let U be an affine open neighbourhood of x . Let $L^\bullet$ be a projective resolution of the module $F(U)$, where the L^i are of finite type. By hypothesis, the ring $\mathcal{O}_{X,x}$ is regular, hence the projective dimension of F_x is finite; let d be this integer. Let

$$K = \ker(L^{-d} \longrightarrow L^{-d+1}).$$

The module K_x is free, for d is the projective dimension of F_x ([M], Ch. VI. Prop. 2.1). By (EGA 0_I 5.4.1 Errata), one deduces that the $\mathcal{O}_U$ -Module $\widetilde{K}$ is free on a neighbourhood U'

⁶**N.D.E.** *stricto sensu*, this is a corollary of the proof which follows and not of the statement. The implication c) $\Rightarrow$ a) is tautological. The other direction is not, but results from the proof. Let us be precise. As below, one covers X by opens immersible in regular schemes, which allows one as explained below to reduce to $X = \text{Spec}(A)$ affine regular and $F = \widetilde{M}$ where M is an A -module of finite projective dimension. It is shown in this case that the conditions a) and c) are equivalent to the dual conditions a') and c'). One then shows that c') implies question d) (see *infra*) which itself implies a'). See the considerations following 2.4.

⁷**N.D.E.** this condition was only in the body of the proof and not in the statement of the corollary; as it is used in 3, it has been added.

⁸**N.D.E.** in fact, a) for X implies a) for all the X_j as asserted in the original text, but for this one must read the proof which follows in detail. This implication does not seem formal at this stage. Let us indeed denote by an index J the conjunctions of a property a), b) or c) for the X_j . It is proved in the proof *infra* that $c_J \Rightarrow a_J$ (this is the chain of implications c') $\Rightarrow$ d) $\Rightarrow$ a'). Now, one has tautologically a) $\Rightarrow$ c), and c) $\Leftrightarrow c_J$, whence a) $\Rightarrow a_J$.

of x , $U' \subset U$. Choosing $f \in \mathcal{O}_X(U)$ such that $x \in D(f) \subset U'$, one thus has a projective resolution of $M_{f'}$ ($M = F(U)$):

$$0 \longrightarrow K_f \longrightarrow (L^{d-1})_f \longrightarrow \cdots \longrightarrow M_f \longrightarrow 0,$$

which proves that the function under study is upper semicontinuous. Now X is quasi-compact, whence the conclusion.

We henceforth suppose X affine noetherian regular and we suppose that $F = \widetilde{M}$, where M is an A -module of finite type, necessarily of finite projective dimension. We shall proceed in several steps. First, we find a condition d), equivalent to a), and prove that it is likewise equivalent to c). Then, with the help of the spectral sequence of the preceding number, we prove d) $\Rightarrow$ b). It remains then to prove that (iii) $\Rightarrow$ (ii); indeed, (i) $\Leftrightarrow$ (ii) $\Rightarrow$ (iii) results immediately from a) $\Leftrightarrow$ c) $\Rightarrow$ b).

Let $x \in U$; by hypothesis $\mathcal{O}_{X,x}$ is a regular local ring; denoting by D the dualizing functor relative to the local ring $\mathcal{O}_{X,x}$, it follows from (V V.2.1) that:

$$D H^{i-c(x)}(F_x) \approx \text{Ext}_{\mathcal{O}_{X,x}}^{d(x)-i}(F_x, \mathcal{O}_{X,x}),$$

where one has set

$$(2.2) \quad d(x) = \dim \mathcal{O}_{X,x} + c(x) = \dim \mathcal{O}_{X,x} + \text{codim}(\overline{\{x\}} \cap Y, \overline{\{x\}}).$$

Now X is noetherian and F coherent, hence:

$$(2.3) \quad D H^{i-c(x)}(F_x) \approx (\underline{\text{Ext}}_{\mathcal{O}_X}^{d(x)-i}(F, \mathcal{O}_X))_x.$$

Moreover, for a module to be zero, it is necessary and sufficient that its dual be so (cf. the editor's note (dualnul) of page dualnul). For every $q \in \mathbf{Z}$, set:

$$(2.4) \quad \begin{cases} S_q = \text{Supp } \underline{\text{Ext}}_{\mathcal{O}_X}^q(F, \mathcal{O}_X), \\ S'_q = S_q \cap U, (U = X - Y), \\ Z_q = \overline{S'_q} \cap Y. \end{cases}$$

From formula (2.3), it follows that a) and c) are respectively equivalent to

a') for every $q \in \mathbf{Z}$ and every $x \in S'_q$, one has $q + i \neq d(x)$.

c') for every $q \in \mathbf{Z}$ and every $x \in S'_q$, if $c(x) = 1$, one has $q + i \neq d(x)$.

Here is the condition d) promised above:

d) for every $q \in \mathbf{Z}$ and every $y \in Z_q$, one has $q + i \neq \dim \mathcal{O}_{X,y}$.

These conditions are equivalent:

a') $\Rightarrow$ c') for the record.

d) $\Rightarrow$ a'). Indeed, let $q \in \mathbf{Z}$ and let $x \in S'_q$; let $y \in \overline{\{x\}} \cap Y$ which⁹ "gives the codimension", i.e. such that:

$$(2.5) \quad \dim \mathcal{O}_{\overline{\{x\}},y} = \text{codim}(\overline{\{x\}} \cap Y, \overline{\{x\}}) = c(x).$$

From the fact that X is regular at y , one deduces:

$$(2.6) \quad \dim \mathcal{O}_{X,y} = d(x) \quad (\text{cf. (2.2)}).$$

But $y \in \overline{\{x\}}$, hence $y \in Z_q$, whence the conclusion.

c') $\Rightarrow$ d). Let $q \in \mathbf{Z}$ and let $y \in Z_q$. Let us admit provisionally that there exists $x \in S'_q$ such that:

$$y \in \overline{\{x\}} \text{ and } \dim \mathcal{O}_{\overline{\{x\}},y} = 1;$$

(one also says that x follows y). It follows that $c(x) = 1$, for y "gives the codimension of $\overline{\{x\}} \cap Y$ in $\overline{\{x\}}$ ", since $x \notin Y$. By c') one draws from this

$$q + i \neq d(x).$$

⁹**N.D.E.** in all that follows, the closures of points are endowed with the reduced structure.

Whence the conclusion, if one remarks that $d(x) = \dim \mathcal{O}_{X,y}$ (2.6). The result admitted is expressed in the following lemma:

LEMMA 2.5. *Let X be a locally noetherian prescheme and let Y be a closed subset of X . Set $U = X - Y$ and suppose that U is dense in X . For every $y \in Y$, there exists $x \in U$ “which follows it”, i.e. such that:*

$$y \in \overline{\{x\}} \text{ and } \dim \mathcal{O}_{\overline{\{x\}},y} = 1.$$

We have applied the lemma by taking for X the prescheme $\overline{S'_q}$ and for Y the subset $Y \cap \overline{S'_q}$.

PROOF OF 2.5. There exists $x \in U$ such that $y \in \overline{\{x\}}$; let us therefore choose an $x \in U$ such that $y \in \overline{\{x\}}$ and such that $\dim \mathcal{O}_{\overline{\{x\}},y} = r$ is minimal. One must prove that $r = 1$. Since one has chosen x in such a way that every $z \in \text{Spec}(\mathcal{O}_{\overline{\{x\}},y})$, $z \neq x$, lies in Y , $\{x\}$ is open in $\text{Spec}(\mathcal{O}_{\overline{\{x\}},y})$. Whence the conclusion. $\square$

The second step consists in deducing b) from d).

Set $D(Z_q) = \{\dim \mathcal{O}_{X,y} \mid y \in Z_q\}$. By d), one knows that, for every $q \in \mathbf{Z}$, one has $q + i \notin D(Z_q)$. One then applies VII.VII.2.3, and one sees that

$$\underline{\text{Ext}}_Y^{q+i}(\underline{\text{Ext}}^q(\mathcal{F}, \mathcal{O}_X), \mathcal{O}_X) \text{ is coherent.}$$

The initial term of the spectral sequence of the preceding number is given by:

$$E_2^{p,q} = \underline{\text{Ext}}_Y^p(\underline{\text{Ext}}^{-q}(\mathcal{F}, \mathcal{O}_X), \mathcal{O}_X).$$

One deduces that $E_2^{p,q}$ is coherent for every $p \in \mathbf{Z}$ and every $q \in \mathbf{Z}$ such that $p + q = i$. Now there are only finitely many pairs (p, q) such that $p + q = i$, and this spectral sequence converges to $\underline{H}_Y^*(\mathcal{F})$, whence the conclusion.

It remains for us to prove that (iii) $\Rightarrow$ (ii). Denote:

$$i: U \longrightarrow X$$

the canonical immersion of U in X . Taking into account the homology exact sequence of the closed set Y (I I.2.11) one sees that (iii) *is equivalent to:*

(iv) $R^i i_*(\mathcal{F}|_U)$ is coherent for $i < n$.

Indeed, one has an exact sequence:

$$0 \longrightarrow \underline{H}_Y^0(\mathcal{F}) \longrightarrow \mathcal{F} \longrightarrow i_*(\mathcal{F}|_U) \longrightarrow \underline{H}_Y^1(\mathcal{F}) \longrightarrow 0.$$

Now $\underline{H}_Y^0(\mathcal{F})$ is a quasi-coherent subsheaf of $\mathcal{F}$ which is coherent, hence is coherent. Hence $\underline{H}_Y^1(\mathcal{F})$ is coherent if and only if $i_*(\mathcal{F}|_U)$ is. Moreover, if $p > 0$, the cohomology exact sequence of the closed set Y reduces to isomorphisms:

$$R^p i_*(\mathcal{F}|_U) \xrightarrow{\sim} \underline{H}_Y^{p+1}(\mathcal{F}).$$

We are going to prove that (iv) $\Rightarrow$ (ii). For this, let us recall (ii):

(ii) for every $x \in U$ such that $c(x) = 1$, one has $\text{prof } \mathcal{F}_x \geq n$.

We argue by induction on n .

If $n = 0$, both conditions are empty.

If $n = 1$, one supposes that $i_*(\mathcal{F}|_U)$ is coherent. Let us argue by contradiction and suppose that there exists $x \in U$ such that $c(x) = 1$ and $\text{prof } \mathcal{F}_x = 0$, i.e. $x \in \text{Ass } \mathcal{F}_x$. Let $y \in \overline{\{x\}} \cap Y$ such that $\dim \mathcal{O}_{\overline{\{x\}},y} = 1$. Set:

$$A = \mathcal{O}_{X,y} \text{ and } X' = \text{Spec}(A).$$

Let us perform the change of base $v: X' \rightarrow X$, which is flat.

$$(2.7) \quad \begin{array}{ccc} U' = X' \times_X U & \xrightarrow{v'} & U \\ i' \downarrow & & \downarrow i \\ X' & \xrightarrow{v} & X. \end{array}$$

The morphism i is separated (for it is an immersion), and of finite type (for it is an open immersion and X is locally noetherian), the change of base is flat hence (EGA III 1.4.15) one has an isomorphism:

$$(2.8) \quad v^*(i_*(F|_U)) \approx i'_*(v'^*(F|_U)).$$

Denote by $\underline{x}$ (resp. $\underline{y}$) the ideal of A corresponding to x (resp. y).

Set $G = v'^*(F|_U)$; G is coherent and $\underline{x} \in \text{Ass } G$, hence there exists a monomorphism $\mathcal{O}_{\overline{\{x\}}} \rightarrow G$, and consequently $i'_*(\mathcal{O}_{\overline{\{x\}}}|_{U'})$ is coherent. By the choice of y , $\dim A/\underline{x} = 1$, and consequently the support of $\mathcal{O}_{\overline{\{x\}}}$ is reduced to $\overline{\{x\}} = \{x\} \cup \{y\}$, for $\overline{\{x\}} = \text{Spec}(A/\underline{x})$ as a scheme. It follows that

$$(\mathcal{O}_{\overline{\{x\}}}|_{U'})(U') = \text{Frac}(A/\underline{x}),$$

ring of fractions of $A/\underline{x}$, and

$$i'_*(\mathcal{O}_{\overline{\{x\}}}|_{U'})(X') = \text{Frac}(A/\underline{x}).$$

But $\text{Frac}(A/\underline{x})$ is not an A -module of finite type for $\underline{x}$ is different from the maximal ideal of A . Whence a contradiction.

Suppose $n > 1$ and the result acquired for the $n' < n$. By the induction hypothesis, for every $x \in U$ such that $c(x) = 1$, one has $x \notin \text{Ass } F_x$. Let such an x , and let $y \in \overline{\{x\}} \cap Y$ such that x follows y , i.e. $\dim \mathcal{O}_{\overline{\{x\}}, y} = 1$. One performs the change of base $v: \text{Spec}(\mathcal{O}_{X, y}) \rightarrow X$, retaining the notations of diagram (2.7). One finds, applying (EGA III 1.4.15), isomorphisms:

$$v^*(R^p i_*(F|_U)) \simeq R^p i'_*(v'^*(F|_U)), \quad p \in \mathbf{Z}.$$

One thus reduces to the case where X is the spectrum of a local ring A in which $\underline{x}$ is a prime ideal of dimension 1, i.e. $\dim A/\underline{x} = 1$. Let us then set $F' = \underline{\Gamma}_Y(F)$ and $F'' = F/F'$. One sees that $F_x \simeq F''_x$ and that $y \notin \text{Ass } F''$. Moreover $F'|_U = 0$ whence, by the exact sequence of the $R^p i_*$, isomorphisms:

$$R^p i_*(F|_U) \simeq R^p i_*(F''|_U), \quad p \in \mathbf{Z}.$$

Since $n > 1$, one deduces that neither x nor y belongs to $\text{Ass } F''$. Now, $\underline{x}, \underline{y}$ are the only prime ideals of A which contain $\underline{x}$; it follows (III III.2.1) that there exists an element $g \in \underline{x}$ which is M -regular, where one has set $F = \widetilde{M}$, $M = F(X)$. Whence an exact sequence:

$$0 \longrightarrow M \xrightarrow{g'} M \longrightarrow N \longrightarrow 0,$$

in which g' denotes the homothety of ratio g in M . By the homology exact sequence, one sees that:

$$R^p i_*(\widetilde{N}|_U) \text{ is coherent for } p < n - 1,$$

hence, by the induction hypothesis, $\text{prof}(\widetilde{N})_x \geq n - 1$, hence $\text{prof } F_x \geq n$, □

3. Applications

One deduces from these results a coherence condition for the higher direct images of a coherent sheaf by a *morphism which is not proper*.

THEOREM 3.1. *Let $f: X \rightarrow Y$ be a morphism of preschemes. Suppose that Y is locally noetherian and that f is proper. Suppose that X is locally immersible in a regular prescheme. Let $n \in \mathbf{Z}$. Let U be an open of X and let F be a coherent $\mathcal{O}_U$ -Module. Suppose that, for every*

$x \in U$ such that $\text{codim}(\overline{\{x\}} \cap (X - U), \overline{\{x\}}) = 1$, one has $\text{prof } F_x \geq n$. Then the $\mathcal{O}_Y$ -Modules $R^p(f \circ g)_*(F)$ are coherent for $p < n$, where g is the canonical immersion of U in X .

Indeed, there exists a Leray spectral sequence whose abutment is $R^*(f \circ g)_*(F)$ and whose initial term is given by:

$$E_2^{p,q} = R^p f_*(R^q g_*(F)).$$

Moreover, there exists a coherent $\mathcal{O}_X$ -Module G such that $G|_U \simeq F$, (EGA I 9.4.3). It then follows from the preceding paragraph that condition (iv) of page 6 is satisfied, i.e. that $R^q g_*(G|_U)$ is coherent for $q < n$. One then applies the finiteness theorem of EGA III 3.2.1 to f and to the sheaves $R^q g_*(F)$, and one finds that $E_2^{p,q}$ is coherent for $q < n$, whence the conclusion.

PROPOSITION 3.2. *Let X be a locally noetherian prescheme locally immersible in a regular prescheme. Let U be an open of X and let $i: U \rightarrow X$ be the canonical immersion. Let $n \in \mathbf{Z}$. Let finally F be a coherent $\mathcal{O}_U$ -Module which is Cohen-Macaulay (on U !). The following conditions are equivalent:*

- (a) $R^p i_*(F)$ is coherent for $p < n$;
- (b) for every irreducible component S' of the closure $\overline{S}$ of the support S of F , one has:

$$\text{codim}(S' \cap (X - U), S') > n.$$

Let G be a coherent $\mathcal{O}_X$ -Module such that $G|_U \simeq F$ (EGA I 9.4.3). Applying corollary 2.3 to G , one finds that condition (a) is equivalent to (c):

- (c) for every $x \in S$, one has $\text{prof } F_x > n - c(x)$, with

$$c(x) = \text{codim}(\overline{\{x\}} \cap Y, \overline{\{x\}}).$$

(a) $\Rightarrow$ (b). Indeed, let S' be an irreducible component of $\overline{S}$ and let s be its generic point. Since F is Cohen-Macaulay, one has $\text{prof } F_s = \dim \mathcal{O}_{S,s} = 0$. Moreover, $\overline{\{s\}} = S'$, whence the conclusion.

(b) $\Rightarrow$ (a). Let $x \in S$ and let S' be an irreducible component of $\overline{S}$ such that $x \in S'$. Let s be the generic point of S' . Since F is Cohen-Macaulay, one knows that:

$$\text{prof } F_x = \dim \mathcal{O}_{\overline{\{s\}}, x}.$$

If $c(x) = +\infty$, there is nothing to prove. Otherwise, there exists $y \in Y \cap \overline{\{x\}}$, such that:

$$c(x) = \dim \mathcal{O}_{\overline{\{x\}}, y}.$$

Now $\mathcal{O}_{X,y}$ is a quotient of a regular local ring by hypothesis, hence:

$$\dim \mathcal{O}_{\overline{\{s\}}, y} = \dim \mathcal{O}_{\overline{\{s\}}, x} + \dim \mathcal{O}_{\overline{\{x\}}, y} > n.$$

Q.E.D.

Bibliography

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